

Niloufar Modirshanehchi

(+49) 15757941944 · Berlin, Germany · n.m.shanechi@gmail.com
github.com/PersianSpock · linkedin.com/in/nil-modirshanehchi

Education

B.Sc. in Computer Science

Sep. 2025 – present

Gisma University of Applied Sciences, Germany

- Pursuing a second B.Sc.

Associate's degree in Electrical Engineering

2018 – 2023

Ferdowsi University of Mashhad, Iran

- Left the B.Sc. degree.

Skills

Programming Languages Python, SQL, C++

Tools & Libraries Pandas, NumPy, PyTorch, TensorFlow, OpenCV, Scikit-Learn, LLM APIs, AWS (S3, SageMaker, EC2), Git, Docker

Data Analytics ETL/ELT Pipelines (Airflow, dbt), Data Warehousing (Snowflake, BigQuery), Data Lake Architectures, A/B Test Evaluation, User Behavior Analysis, Financial & Operational Metric Analysis, Visualization Tools (Power BI, Tableau, Python libraries)

Machine Learning NLP Pipelines, LLM-based NLP Workflows, Transformer Models (Hugging Face), LLM Applications (RAG, API Integration), Recommender Systems, Feature Engineering, Model Evaluation

Natural Languages English (C1), German (A1 – attending B1 intensive course)

Technical Experience

Werkstudent Data Engineering

May 2026 – present

thekey.academy (Working student)

Berlin, Germany

- Working on analytics infrastructure and tracking quality across web, Android, and iOS platforms using BigQuery and GA4.
- Supporting data pipelines for ad platform data (Google Ads, Meta Ads) and cross-platform event tracking.

Junior Data Scientist

Jul 2023 – Sep 2023

Hotelian.com (Full-time)

Dubai, UAE

- Contributed to the development of a conversational AI platform for hotel bookings, presented at ITB Asia 2023, which focused on LLM-based content understanding, personalized recommendations, and search relevance.
- Built Data and NLP pipelines in Python and SQL over 1M+ user reviews, hotel metadata, and behavioral signals to learn reusable representations; replacing heuristic baseline with embedding-based ranking model, improving offline AUC from 0.68 → 0.75 (+10% relative uplift) on user-level holdout; validated via a 2-week A/B test (CTR +4.7%, $p < 0.05$).
- Developed a personalized ranker using semantic embeddings, Elasticsearch, and GIS features; improving offline ranking (NDCG@10) from 0.31 to 0.37 (+19% relative) vs. the prior production ranker on a time-based validation set.
- Designed production KPIs and model monitoring (data quality, drift signals, conversion, latency, error rates) to track impact and reliability during rollouts.

Remote Research Intern

Aug 2022 – Jul 2023

University College London (Internship)

London, UK

- Worked on the core recommendation engine for personalization and ranking systems on a UCL-backed startup.

- Built a SQL feature extraction pipeline, and implemented contextual bandit (ϵ -greedy) for ranking optimization.
- Improved simulated NDCG@10 from 0.42 to 0.45 (+7% relative) vs. an implicit MF baseline using offline replay evaluation.
- Produced end-to-end ML workflows on AWS (EC2, S3, SageMaker), reducing model iteration/training time by 10%.
- Supported product decisions through model-driven content ranking experiments and A/B tests, and tracking conversion KPIs plus basic drift/data-quality checks after rollout.

Junior Data Analyst & NLP Researcher

July 2020 – May 2023

Wise Intelligent Agents (ahdsoft) (Part-time)

Mashhad, Iran

NLP Researcher:

- Built and deployed multiple NLP systems using pretrained transformer models (BERT, T5, TrOCR), leading supervised fine-tuning across OCR, ASR, and text classification tasks in low-resource Farsi, including CTC-based wav2vec2 adaptation and domain-specific tokenizer adaptation.
- Developed shared fine-tuning and evaluation pipelines for Farsi ASR, combining academic datasets (e.g., Common Voice) with large-scale proprietary data (recorded noisy conversations in Farsi); reducing WER by 25–30% (relative) vs. prior production models via systematic fine-tuning and data augmentation.

Junior Data Analyst:

- Delivered 150+ dashboards tracking revenue, cost, conversion, and operational KPIs across logistics, marketing, and sales, enabling data-driven decisions in budgeting, pricing, and operational planning.
- Developed and maintained ETL pipelines to transform raw data from data lake into structured warehouse schemas, improving data reliability and reporting efficiency, reducing query time and storage costs.
- Performed EDA workflows, customer segmentation, and user-behavior analysis for supporting targeted marketing and content strategies, leading to conversion improvements.
- Collaborated with product and business stakeholders to define KPIs and perform analysis on trade-offs between growth, customer acquisition efficiency, and operational costs.

Remote Research Intern

July 2021 – Sep. 2021

Dublin City University (Internship)

Dublin, Ireland

- Contributed to a research project on Graph Neural Networks (GNNs) aimed at extracting structured knowledge from deep neural networks and generating human-readable local explanations.
- Developed and implemented semantic knowledge graph models using Neo4j while utilizing Python, PyTorch, and TensorFlow on CIFAR-10 and CIFAR-100 datasets.
- Supported research efforts through practical ML implementations, laying a strong foundation in advanced neural network techniques.

Activities

- Olympiad teacher for high school students (Combinatorics, Advanced Algorithms, Graph Theory) — 2018–2021.
- Problem Setter & Judge at 16th International ISC Conference on Information Security and Cryptology (ISCISC'19) — 2019.
- Problem Setter & Judge for state-wide Mathematics Olympiad for middle school students — 2018.